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THIS FILE INCLUDE ALL DATA OF PREVIOUS FILES LIKE QUIZ ,GRAND QUIZ ,SHORT AND LONG QUESTIONS.	FINAL TERM

Point Mutation?

A **point mutation** or substitution is a genetic **mutation** where a single nucleotide base is changed, inserted or deleted from a sequence of DNA or RNA. **Point mutations** have a variety of effects on the downstream protein product—consequences that are moderately predictable based upon the specifics of the **mutation**.

Complications Of Inheritance?

Complications to inheritance patterns: Multiple genes - epistasis, polygenic traits Genes & the environment - sex-influenced traits, incomplete penetrance.

Complications : A common recessive inheritnace can give dominant pattern Autosomal dominant inheritance with non-penetrance, Autosomal dominant inheritance with variable expression.

Mutation Effects In Functioning Of Genes?

Some mutations don't have any noticeable effect on the phenotype of an organism. This can happen in many situations: perhaps the mutation occurs in a stretch of DNA with no function, or perhaps the mutation occurs in a **protein**-coding region, but ends up not affecting the amino acid sequence of the **protein**.

Causes Of Epigenetics?

Stress Hormone **Causes Epigenetic** Changes. Researchers found that chronic exposure to a stress hormone **causes** modifications to DNA in the brains of mice, prompting changes in gene expression

Function Of Pedigree?

A pedigree may be used to establish the probability of a child having a particular disorder or condition. It may be used to discover where the genes in question are located (x, y, or autosome chromosome), and to determine whether a trait is **dominant** or recessive.

Define Epigenetics?

Epigenetics is the study of heritable phenotype changes that do not involve alterations in the DNA sequence

RAPD And ALFP Difference?

RAPD is Random Amplification of Polymorphic DNA'. It is a type of PCR, but the segments of DNA that are amplified are random.

AFLP is a PCR-based tool used **in genetics** research, DNA fingerprinting, and **in**

the practice of **genetic** engineering. Developed **in** the early 1990s by Keygene, **AFLP** uses restriction enzymes to digest genomic DNA, followed by ligation of adaptors to the sticky ends of the restriction fragments.

Forward, Reverse, And Neutral Mutations: 3marks

Forward mutation: Mutation that alters the wild type phenotype into mutant phenotype.
Reverse mutation: Mutation that changes a mutant phenotype back into the wild type.
Neutral mutation: Mutation that alters the amino acid sequence of the protein but does not change its function as replaced amino acid is chemically similar or the affected AA has little influence on protein function

Point Mutations And Enlist 3 Structural Changes In Chromosomes?

A **point mutation** or substitution is a genetic **mutation** where a single nucleotide base is changed, inserted or deleted from a sequence of DNA or RNA. **Point mutations** have a variety of effects on the downstream protein product—consequences that are moderately predictable based upon the specifics of the **mutation**.
Deletion, duplication, translocation, transcription.

Structural Change in the Structure of Chromosomes. ... Chromosomes are the vehicle of hereditary material or genes. Any alteration, addition or deletion of chromosomal part leads to alteration of number, position or sequence of genes in the chromosome.

Complications In Inheritance?

Epistasis, polygenic traits Genes & the environment - sex-influenced traits, incomplete penetrance.

Modification Of Epigenetica

Epigenetic modification - non coding RNA is a new mechanism for gene regulation. RNA which is not used for making proteins can be cleaved and used to inhibit protein-coding RNAs (siRNAs, microRNAs). Three common types of modification - Cytosine methylation Histones modifications siRNAs

Mutation Alter Genes Expression?

Non-coding mutations: Promoter/enhancer element trinucleotide repeats in 5 and 3'UTRs Splice sites. Effect of mutation on gene function. Null allele (loss of function) - no gene product.
Hypomorph – decreased activity. Gain of function - increased activity. Dominant negative-

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antagonizes normal product. Neomorph - novel activity. Predicting that a gene product won't do the job. Deletion, nonsense, frameshift of sequence is deleterious. Mutation in splice site usually has sever effects. Missense mutations depend on location in protein. Is it non-conservative. Is AA conserved in evolution.

RFLP And RAPD?

RFLP is a technique that exploits variations in homologous DNA sequences. **RFLP** analysis was an important tool in genome mapping, localization of genes for genetic disorders, determination of risk for disease, and paternity testing.

RAPD is Random Amplification of Polymorphic DNA'. It is a type of PCR, but the segments of DNA that are amplified are random.

Genetic Mapping?

Gene mapping describes the methods used to identify the locus of a gene and the distances between genes. The essence of all genome mapping is to place a collection of molecular markers onto their respective positions on the genome.

Sickle cell anemia?

Sickle cell anemia (**sickle cell disease**) is a disorder of the blood caused by an inherited abnormal hemoglobin (the oxygen-carrying protein within the red blood **cells**). The abnormal hemoglobin causes distorted (sickled) red blood **cells**. The sickled red blood **cells** are fragile and prone to rupture.

Point mutation and cause?

A **point mutation** or substitution is a genetic **mutation** where a single nucleotide base is changed, inserted or deleted from a sequence of DNA or RNA. **Point mutations** have a variety of effects on the downstream protein product—consequences that are moderately predictable based upon the specifics of the **mutation**.

Cause:

The protein may lose its function, which can **result in** a disease in the organism. For example, sickle-cell disease is **caused** by a single **point mutation** (a missense **mutation**) in the beta-hemoglobin gene that converts a GAG codon into GUG, which encodes the amino acid valine rather than glutamic acid.

Mutation nomenclature?

Standard **nomenclature** based on genomic DNA reference sequences and RNA reference sequences is not shown. "Genomic DNA reference sequence" simply indicates any human DNA sequence in the database that is not based on a cDNA sequence. Example of standard

mutation nomenclature based on a coding DNA sequence.

VNTR and application?

VNTR are short nucleotide sequences organized as a tandem repeats on genomes.Found on many chromosomes, Show variations in length between individuals. **Applications of VNTR** Microbiology DNA fingerprinting Genetic diversity Forensics Mapping of genomes Breeding patterns of wild or domesticated animals .

Single And Multifactorial Mutation?

Single gene disorders are caused by DNA changes in one particular **gene**, and often have predictable inheritance patterns. Over 10,000 human disorders are caused by a change, known as a **mutation**[?], in a **single gene**[?]. These are known as **single gene** disorders

Multifactorial Disorders. Researchers are learning that nearly all conditions and **diseases** have a genetic component. Some disorders, such as sickle cell disease and cystic fibrosis, are caused by **mutations** in a single gene. Conditions caused by many contributing factors are called complex or **multi factorial** disorders.

5 Complication Inheritance Pattern?

Complications to inheritance patterns Multiple genes - epistasis, polygenic traits Genes & the environment - sex-influenced traits, incomplete penetrance

Autosomal Domenet Disordered?

A single abnormal gene on one of the first 22 nonsex (**autosomal**) chromosomes from either parent can cause an **autosomal** disorder. **Dominant** inheritance means an abnormal gene from one parent can cause disease. ... Examples of **autosomal dominant disorders** include Marfan syndrome and neurofibromatosis type 1.

Function Of STR?

Short Tandem Repeat (**STR**) analysis is one of the most useful methods in molecular **biology** which is used to compare specific loci on DNA from two or more samples. A short tandem repeat is a microsatellite, consisting of a unit of two to thirteen nucleotides repeated hundreds of times in a row on the DNA strand.

During DNA Extraction How Proteins Are Denatured?

Adding proteinase

Define Locus?

A **locus** (plural **loci**) in **genetics** is a fixed position on a chromosome, like the position of a gene or a marker (**genetic** marker). Each chromosome carries many genes; human's estimated 'haploid' protein coding genes are 19,000–20,000, on the 23 different chromosomes.

Sex Linked Traits And Disorder?

Genes on the X or Y chromosome are called **sex-linked**. ... For example, an X-**linked** recessive allele in humans causes haemophilia, which is much more common in males than females because males are hemizygous and therefore express the **trait** when they inherit one mutant allele

Define Multiple Alleles With Example?

Multiple alleles Three or more alternative forms of a gene (**alleles**) that can occupy the same locus. However, only two of the **alleles** can be present in a single organism. For example, the ABO system of blood groups is controlled by three **alleles**, only two of which are present in an individual.

Genetic Disorder?

A **genetic disorder** is a **genetic** problem caused by one or more abnormalities in the genome. Most **genetic disorders** are quite rare and affect one person in every several thousands or millions. **Genetic disorders** may be hereditary, meaning that they are passed down from the parents' **genes**.

Genetic Mapping?

Gene mapping describes the methods used to identify the locus of a gene and the distances between genes. The essence of all genome mapping is to place a collection of molecular markers onto their respective positions on the genome.

What Is Location Of Intronic Sequence With Respect To Exons?

Sequences that are joined together in the final mature RNA after RNA splicing are **exons**. **Introns** are **found** in the genes of most organisms and many viruses, and can be **located** in a wide range of genes, including those that generate proteins, ribosomal RNA (rRNA), and transfer RNA (tRNA).

Write Down At Least Three Mitochondrial Diseases?

MELAS

Mitochondrial encephalomyopathy, lactic acidosis and stroke-like episodes is MELAS. Kearns-Sayre Syndrome

NARP

NARP stands for neurogenic weakness, ataxia and retinitis pigmentosa.

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Structul chamge in chromosome?

Structural Change in the **Structure** of **Chromosomes**. ... **Chromosomes** are the vehicle of hereditary material or genes. Any alteration, addition or deletion of **chromosomal** part leads to alteration of number, position or sequence of genes in the **chromosome**.

•Define Epistasis

Epistasis and modifier genes Epistasis is a phenomenon that consists of the effect of one gene being dependent on the presence of one or more modifier genes Type of polygenic inheritance where alleles at one gene locus can hide or prevent expression of alleles at second gene locus.

•Genomic Imprinting

Genomic Imprinting Epigenetic marking of a locus on the basis of parental origin. Results in monoallelic gene expression. Some genes are expressed only from the maternal genome and some only from the paternal genome. It is estimated that about 40 genes are imprinted and they can be found on several different chromosomes .

Point mutations

•Point mutations affect single sites on DNA.

•Substitution of one base for another.

•Deletion/addition of a single base or small number of bases.

Transitions

- If purines (A/G) is replaced by purines and pyrimidines (T/C) is replaced by pyrimidines.

Transversion

- If purine substitutes for pyrimidines or pyrimidine is replaced by purines. It is called as transversions.

Result of point mutations

- Silent mutations
- Missense mutations
- Nonsense mutations

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- Frameshift mutations
  result in all amino acids downstream from the mutation site being completely different from wild type. These proteins are generally nonfunctional.

Mutation - Forward, Reverse and Neutral Forward mutation

- Mutation that changes the wild type phenotype into mutant type phenotype.

- Any mutation which changes a functional wild genotype or wild phenotype to nonfunctional form.

Wild type phenotype/ genotype

— **reverse mutation**	Most common phenotype in natural populations of the organism is called a wild type phenotype.
— **neutral mutation**	Mutation that changes a mutant phenotype back into the wild type.
—	Mutation that alters the amino acid sequence of the protein but does not change its function as replaced amino acid is chemically similar or the affected amino acid has little influence on protein function.

Common X-linked dominant diseases

- Rett syndrome
- Alport syndrome
- Goltz syndrome

- Fragile-X Syndrome

Short Tandem Repeat Polymorphism

Short tandem repeat sequences

- STR are repeats of nucleotide sequences.
- AAAAAA… mononucleotide
- ATATAT… dinucleotide
- TAGTAGTAG…trinucleotide
- TAGTTAGTTAGT… tetranucleotide
- TAGGCTAGGCTAGGC… pentanucleotide
- Different alleles contain different numbers of repeats.

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Frameshift mutations

result in all amino acids downstream from the mutation site being completely different from wild type. These proteins are generally nonfunctional.

Common mitochondrial diseases

- Myoclonic epilepsy
- Mitochondrial recessive ataxia
- Leber hereditary optic neuropathy
- Sensory ataxia
  neuropathy 17.

Chiasmata formation —

Chiasmata visible on chromosomes are regions of crossing over. — Occurs between

non-sister chromatids

Crossing over

is the swapping of genetic material that occurs in the germ line. During the formation of egg and sperm cells, also known as meiosis, paired chromosomes from each parent align so that similar DNA sequences from the paired chromosomes cross **over** one another.

During meiosis I, homologous chromosomes separate. During meiosis II, sister chromatids separate.

Prophase I

— Nucleolus disappears. — Meiotic spindle forms. — Nuclear membrane disappears.

Spermatogenesis and oogenesis

Meiosis in males is called spermatogenesis and produces sperms.
Meiosis in females is called oogenesis and produces ova.

— Forward genetics

Not knowing anything about the location of the gene.
— Starting at the phenotype, determining which protein was involved and then getting to the gene through the protein.

reverse genetics.

Starting from the approximate location of the gene, to finding the gene itself.
— Translating it to learn about the protein and its function
Positional cloning or reverse genetics.

nondisjunction chromosome

Nondisjunction occurs when homologous chromosomes (**meiosis** I) or **sister chromatids** (**meiosis II**) fail to separate during **meiosis**

Types of Non-disjunction

Meiotic non-disjunction
Mitotic non-disjunction

Meiotic Non-disjunction If non-disjunction occurs during 1st meiotic division then all the sperms/ova derived from primary spermatocyte/oocytes will be abnormal.

Mitotic Non-disjunction Mitotic non-disjunction results in cells with different number of chromosomes. Breaking of the spindle fibers during metaphase or anaphase.If spindle fibers do not form for one of the chromosomes, then during anaphase it will not separate into two chromatids and one cell will have an extra chromatid and the other will be lacking one. Mitotic nondisjunction causes trisomic and monosomic daughter cells to be formed.

SNP: Single Nucleotide Polymorphisms

Haplotype mapping: sets of alleles or DNA sequences can be clustered so that a single SNP can identify many linked SNPs Single nucleotide polymorphisms (SNPs) are

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polymorphisms that are caused by point mutations that give rise to different alleles containing alternative bases at a given position of nucleotide within a locus. Due to their high abundance in the genome, SNPs already serve as the predominant marker type.

FINAL TERM QUIZ (All quiz collected from all previous files)

1. Genes will be switched on when histones Acetylated
2. vorinostat is histone deacetylase inhibitor
3. Exchange of DNA Segments between.chromosomes is known crossing over
4. Control of expression of gene by The expression of other Gene Epistasis
5. Expression of Phenotype of particular allele by number of individuals Expressivity
6. Centromere inverted in on of the following Chromesomal inverron Pericentric inversions
7. Human made Chromosomes is normally written as 46.XY
8. Chrome somal aberrations are classified into 2 types
9. Formation of Plenipotent cells from totipotent cells is called Cellular differentiation
10. Expression of "Dhene type of particular ailele by number Expressivity
11. Addition of a complete set chromosome is named as Euploidy
12. Two genes close together, frequency of crossing over would be Low
13. Large arm of chromosomes is ? arm
14. Colchicine to promoter Polyploidy
15. :Polygenic inheritance is controlled by Multiple genes
16. Fur color in Labrador is controlled by 2 genes
17. cancer, infertility are result of Translocation
18. : Which is following is not chromosomal aberration Translation
19. which is result of frame shift Deletion
20. Which is not true recessive trait Females are more affected
21. : Molecular biology, forensic , genes mapping are application of VNTR
22. About 25000 SNA in human are Non-Synapse
23. Determination of pattern inheritance Pedigree
24. Types of single nucleoide polymorphism that change the sequence of amino acid Non-coding
25. Expression of phenotype Incomplete inheritance
26. Small case for alphabets used for RNA level
27. Frequency of autosomal recessive 2-2.5

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28. Medelian disorder is Single gene
29. Addition mutation is addition of Nucliotide
30. Which is one of chromosomal duplication Retro - transposition
31. Turner syndrome is an example of Monosom
32. Mutation which type a mutant phenotype back into the wild type Reverse mutation
33. A genetic disease or disorder appears as a result of mutations in an individual' DNA
34. Hunter's disease is an example of_ X-linked recessive disorder
35. A condition in which some people with an appropriate genotype fail to express the phenotype is known as the incomplete penetrance
36. Which one of the following is not influenced by the silent mutation? Phenotype
37. Mutation that produce the change in the amino acid in protein product (histidine in for arginine Missense mutation
38. Base analog have the similar structure with standard bases of the DNA
39. Mutation which convert a DNA sequence into the stop codon. Non sense mutation
40. The term Ashkenazi refers to the specific people of following population.Jew population
41. ABO BLOOD group system is an example of multiple alleles
42. Gene maping give adea of gene location.
43. Which of following have single genotypes Blood group AB and O
44. Fruit clor of summer squish plant is an example Dominent epistatis
45. Pepper color is controlled 2 Gene
46. Position depended method of disease identification following Reverse genetics
47. Genetics can be studied by Two ways
48. Example of Mendilian genetics None
49. Condition in which an intermediate phenotype is developed from cross of parental generation. incomplete dominance
50. Condition in which one gene prevent the expression of other gene epitasis
51. chromosome is made up of DNA
52. structure for chromosome stability telomers
53. Broken ends of chromosomes are sticky
54. Centromere is unable to work due to mutation in CDE III
55. XO system of sex is present in grasshopper

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56. prokaryotic cell divides by binary fission
57. Chromosomes become fit in the central line metaphase
58. Haploid cells are produced by mitosis
59. Tetrads align at center metaphase
60. Genes are present on chromosomes first time said by greogr mendal
61. Cross for determination of genotype of an individual test cross
62. In ABO blood group system Alleles A and B show co-dominance
63. If two genes are more close then frequency of cross over will be high
64. In nature variation is Wide Spread.
65. Intermediate are produced in Incomplete Dominance.
66. Alleles at one gene hid/ prevent expression at second locus Epitasis.
67. Lung congestion occurs in Cystic fibrosis.
68. Cell spend 90% of its time in Interphase.
69. Chromosomes are stabilized by Telomere.
70. Nucleosomes folds and form Loops.
71. Which band is not found in Plants G-band.
72. Most common genetic markers Micro satellite SNPs
73. The possible arrangement of the these genes in which eye color is placed between wala the other two.
74. Cross b/w two for finding genome Test Cross
75. Mechanisms of genetics 2
76. Condition in which an organism is with 47 chromosomes with XXX triple X syndromes
77. Histone Deacetylase inhibitor is Vorionstat
78. When organism / or individual with allele for disease is unable to express the trait Incomplete Peiet .
79. What is not chromosomal aberration Translation.
80. Cancer, infertility leukemia are diseases of Translocation.
81. Chromosome 13 trisomy Patau's syndrome
82. Disease due to monosomy Turner's syndrome
83. Cochicine Promotes Polyploidy.

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84. Mutation in each cell 6
85. Mutation that pass onto next generation Green Line.
86. Organism with suppressor mutation is a Double mutant.
87. Point mutation effects single loci on DNA.
88. Effect DNA sequence but not protein: Silent mutation.
89. Mutations in with small letters are used for nucleotide RNA Level.
90. _____ indicates affected residue.
91. SNP are found on coding & non coding region.
92. 25000 SNPS None synonymous.
93. Pedigree helps to determine mode of inheritance.
94. Frequency of autosomal recessive disorder is 2-2.5
95. In X-linked disorder all are true except Female are more affected than male.
96. When gene and environment both contributes Multifactorial.
97. About 25000 SNA in human are Non Synapse.
98. Expression of Phenotype Incomplete Inheritance
99. Mendilian disorder is single gone.
100. Addition mutation is addition of Nucleoside.
101. Types of single nucleoside polymorphism that change the sequence of amino acid Non-Coding.
102. Which mutation convert amino acid into stop codon non sense mutation.
103. Change in chromosome number is aneuploidy.
104. Type of trisomy which extra chromosomes arms derived from different chromosomes Tertiary Truism.
105. Which blood group do not carry any antigen O Blood Group.
106. Cytokines is occurs in Ten Phase.
107. The Physical location of an allele on a gene is Locus.
108. Sex determination chromosomal pattern in fishes is Z W system
109. The larger arm of the chromosome is named as P-arm
110. Medallion inheritance is Polygenic.
111. Genetic marker for a single locus is AFLP
112. Discontinous variation is Permanent.

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113. Which one has larger chromosome Plants.
114. No. of chromosome in Tobacco Plant 48
115. How many types of chromosomes banding 9
116. Histone Protein are basic bamino acid argentine & lysine
117. In Bacteria DNA is unwind by DNA Gyrase
118. DNA is left handed helix
119. Replication in lagging strand is discontinuous
120. Di-allelic gene generate genotype 3
121. DNA duplication caused by Ratio Transposition.
122. Which on is sexual disease Alzheimer's disease.
123. Phenotype depend on both.

GRAND QUIZ BY RAMZAN SOUROO 2021

1.Blood types AB and O have only **one** genotype each

2.Fragile X syndrome is most commonly caused by a **CGG trinucleotide repeat expansion**

3.**Exposure to toxins and diet** are major factors that affect the epigenome

4.Meiosis in males is called **spermatogenesis** and produces sperms.

5.Have a high percentage of **basic** amino acids, which gives histones an overall **positive** charge, while DNA is **negative** charged.

6.ABO blood system Controlled by **a triallelic gene.**

7.According to **blending hypothesis**, genetic material of the parents

mixes during fertilization

8.Generally, karyotypes represent by arranging in the **descending** order of size keeping their centromere in a **straight line.**

9. a point mutation in the **CDE-III sub-domain of CEN DNA** completely eliminates the ability of the centromere to function during chromosome segregation

10.Polygenic inheritance is a condition **when two or more genes influence the expression of one trait.**

11.Mitotic apparatus/ spindles the apparatus of microtubules controls chromosome movement during mitosis

12.**Quantitative genetics** is the study of group hereditary for traits determined by many

genes simultaneously.

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13.**Sign epistasis** occurs when one mutation has the opposite effect when in the presence of another mutation

14.When two mutations together have a smaller effect than expected from their effects when alone,it is called **negative epistasis**.

15.Inherited variations result from the **activity of genes**.

17.There are about **50** loops in the chromosome of E. coli.These loops are highly twisted or supercoiled structure with about **four** million nucleotide pairs.

18.Variations that result from **acquired characteristics** are not acted upon by natural

selection & thus do not affect evolution.

19.Haploid gametes are formed by **Meiosis.**

20.Which statement is not correct about Recessive Epistasis in pigeons.**Pattern is Epistasic to spread color.**

21. Histone modifications are the cause of **Epistasics.**

22.During Recessive Epistasis **Epistatic gene** exerts its affect with **homozygous recessive** genotype.

23.Which one is not regulate gene expression? **Nucleotide modification**

24.Suppose that the distance between two genes on a linkage map is 7 units.The cross-over frequency between the two genes is **7%.**

25.A cell with 10 chromosomes undergoes mitosis.How many daughter cells are created?.....Each daughter cell has .....chromosomes. **2,10**

26.**Sign epistasis** occurs when one mutation has the opposite effect when in the presence of another mutation.

27.Which is correct about the comb shape in chicken? **Four different comb shapes is controlled by two genes found on two different chromosomes.**

28.Group of adjacent nucleotides are joined by **Phosphodiester bond.**

29.How many numbers of allels can generate 3 genotypes? **Genes with 2 allels.**

30.The tendency of two or more than two genes to stay together during inheritance is called **Linkage.**

31.Which one of the following statement is not correct about Homologous chromosomes?**The chromosomes always move to the same location in the cell during division.**

32.**Dependent segregation**: Alleles at the two gene loci segregate together, and are transmitted as a unit.

33.Formation of pluripotent cells from totipotent cells is called **Cellular differentiation**.

34.Fruit color of summer squash is an example of **Dominant epistasis.**

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35.Skin color pigmentation in humans is controlled by at least **three** separately

inherited genes.

36.Cells splits into two? **Cytokinesis**.

37.Name the organism whose first genetic map was made; **Drosophila**.

38.Stage of prophase at which nuclear membrane dissolves; **Stage 2.**

39.Failure in which of the following phenomenon would result in linkage? **Law of independent assortment.**

40.Chargaff's rules for DNA base pairing indicates that **A+T = C+G**.

41.The genome length of Prokaryotic Chromosomes is smallest in **RNA viruses**.

42. Chromatin fibers undergo folding, coiling and supercoiling during **prophase** so that the chromosomes become progressively thicker and smaller and become visible.

43.**Lampbrush Chromosome** First time observed by Flemming in 1882.

44.Genes will be switched ON when **Histones are acetylated**.

45.In life forms the ZZ-ZW chromosomal sex determination,the males have **ZZ chromosomes.**

46.How many numbers of heterozygotes genotypes will be produced from five numbers of genes specific alleles? **Ten**.

47.Which one of the following statement is correct about Epigenetic Modifications? **When histones are acetylated,chromatin is open and genes are potentially active.**

48.Which one of the following statement is correct? **Father with X-linked disorder will transmit the mutant allele to all daughters but to no sons**.

49.Which one of the following does not revert to the Euchromatin stage? **Constitutive heterochromatin.**

50.New phenotypic variations produced when T.H Morgan test cross performed between F1 Dihybrid with double mutant because of --- **crossing over and linkage that break the genes physical connection.**

51.Which of the following organism have 16 chromosomes? **Onion and Mold.**

52.Which one of the following statement is correct about human genes? **70% are mono-allelic; they exist only in one form.**

53.Which one of the following statement is not correct about epigenetic regulation in identical twins? **The epigenetic tags can have such an effect on the twins that one can de**

velop a disease while the other is also.

54.Which one of the following statement is not correct about organism with their chromosomes

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numbers? **Angiosperms have 78 chromosomes.**

55.Who is considered "the Father of Genetics"? **Gregor Mendel.**

56.Which one of the following statement is not correct about Prophase II? **Replication of chromosomes occur.**

57.Which one of the following statement is correct about to the Mendal Law of Dominance? **In a cross of parents, that are pure for contrasting traits, only one form of the trait will appear in the next generation.**

58.Which trait will produce in F1 generation of the crossing a brown pepper (RRyy) with a Yellow pepper rrYY? **Red.**

59.The region where two sister chromatids of a chromosome joined or "held together" during mitotic metaphase is called **centromere**.

60.In metaphase I, **tetrads** line up at the metaphase plate, with one chromosome facing each pole.

61.How many types of enzymes are involved in histone acetylation? **Two.** 62.Which Condition is called as Polygenic inheritance? **When two or more genes influence the expression of one trait.**

63.Each identical side of chromosome is **Chromatid.**

64.In humans, **~ 220 types** of somatic cells.

65.**Angiosperms** have from 8 – 16 (most common number 12).

66.**Mendel s' Law of Independent Assortment** states that the two alleles of a gene found on each of chromosome separate independently of one another into sex cells.

67.Tetrads align at center.....**Metaphase I.**

68. It is estimated that about **40 genes** are imprinted and they can be found on several different chromosomes.

69.Causes genes to not segregate independently.Causes genes to be more likely to be inherited together.Represents an exception to Mendal's Law of Independent Assortment**...Gene linkage.**

70.Which one of the following is widest DNA? **A-DNA.**

71.What is the composition of nucleotide? **A base + A sugur + Phosphate.**

72.What is the name of the region where double-stranded single circular DNA is found in the prokaryotic cell? **Nucleoid**.

73.Shepherd's purse has two shape heart and allow.If you cross a true breeding heart to a true breeding narrow.F1 will give .....**Heart.**

74.In the ABO blood system,blood group 'O' is characterized by the; **Absence of both antigen A and antigen B.**

75.What is the other name of a Polytene chromosome? **Salivary glands chromosomes.**

76.**About 5000** bands can be recognized on a giant chromosome of the Drosophila.

77.Which one of the following histone protein has high Molecular weight,greater number of amino acids and high percentage of arginine and lysine? **H1**

78.During **Cellular differentiation**,**totipotent** cells become **pluripotent** cells of the embryo which

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differentiate into specific lineages.

79.**Continuous variation** describes the situation in which there are many intermediates between the extremes.

80.In blue-eyed Mary plants a white substance is first converted to a **Colorless** substance by 1st gene and that to a **Magenta** substance by 2nd gene to give a blue flower.

81.Technique that gives an rough idea of gene location? **Genetic mapping**.

82.Modern Concept about the Law of dominance is.....**The gene for a particular inherited character resides at a specific locus on homologous chromosome.**

83.Spindle microtubules attach to the kinetochores of chromosomes and move the chromosomes to the **metaphase plate.**

84.Variation can seen among individuals of same species.**It can be due to genetic makeup, environment or by combination of both.**

85.In E.coli, about **3000 to 4000** genes are organized into its one circular chromosome.

86.Building blocks of nucleic acids are **Nucleotide**.

87..Which one of the following sequence shows the correct life cycle of human? **Diploid egg+diploid sperm-->haploid zygote-->mitosis-->haploid adult-->meiosis-->diploid gametes.**

88.Diameter of chromatin is **30 nm.**

89.Which one of the following statement is not correct about incomplete dominance? **One of the allele is dominant.**

90.All offsrings of same variety is called as **True breeding.**

91.When a cell divides,what do we call the 2 new cells? **Daughter cells.**

92.The numbers of genotypes can be calculated on the basis of **Number of alleles.**

93.Which one of the following statement is not correct about the influence of environmental factors on epigenetics? **Social interactions and physical activity not affect the epigenome.**

94. A heredity unit on chromosomes? **Gene.**

95.Which one of the following has longer chromosomes? **Plants.**

96.During prophase,the Nuclear envelope.....**Disappear.**

97. Chromosomes were described by **Strausberger** in 1875.Chromosome was first used by **Waldeyer** in 1888.

98.Type of cell division important for tissue regeneration is **Mitosis.**

99. Genetic make-up of an organism is called **Genotype**.

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99.Physical appearance of an organism is called **Phenotype.**

100.**A physical map** is primarily based on the locations of landmarks along a DNA molecule.

101.The giant chromosomes found in **diptera** and they may be as long as 300 μ and up to 10 μ in diameter.

102.ABO blood group in humans is determined by **Multiple alleles.**

103.How many numbers of alleles control the fur color of rabbit? **Four.**

104.Which one of the following is not the property of Garden pea plant that was used by Gregor Mendel for his experiment? **Long generation time.**

105.Fragile X chromosome affects the **FMR1 gene.**

106.A female bee sex is **ZW.**

107.Epistasis is the interaction between **2 genes**.

108.Which is the Causes of Epigenetics? **(DNA Methylation,Histones modifications,miRNA) All mentioned.**

109.70% Gene in human are **Mono-allelic**.

110. DNA Replication is **Semiconservative.**

111.Due to **crossing-over** during meiosis I, the two sister chromatids of each chromosome are no longer genetically identical.

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